

EE 616 Project Update

Vision Only Leader Follower Coordination on Constrained Edge Hardware

Revised scope and preliminary results plan for faculty review

STATUS

Proposed for review

OWNER

Neeraj Kumar Kanchani

LAST UPDATED

September 15, 2026

Field	Details
Candidate	Neeraj Kumar Kanchani
Faculty reviewer	Professor Masudul Imtiaz
Course	EE 616-57, section 9189, four credits
Target completion	First week of December 2026
Scope	Simulation-only decentralized leader-follower perception and control using ROS 2 Jazzy and Gazebo Harmonic

1 Executive Summary and Review Request

I propose a simulation-only study of decentralized leader-follower coordination on constrained edge hardware. A follower autonomous mobile robot will use a forward-facing RGB camera and its own wheel odometry to estimate the relative state of a leader, maintain a commanded formation, and recover after short visual occlusions. ROS 2 namespaces will isolate each robot. Gazebo ground truth will be recorded for evaluation and will never enter the control path.

This revision narrows my earlier warehouse intersection-negotiation idea, which included LiDAR and stereo fusion. The narrower scope makes the research question measurable within the available semester while preserving the main themes of decentralized perception, edge inference, state estimation, and autonomous control.

Review requested.

I am asking for confirmation that this revised scope is suitable for EE 616 before I proceed beyond the preliminary baseline. The first evidence package will compare monocular range and bearing estimates with simulator ground truth and report latency, tracking error, and GPU memory use.

2 Academic Context and Scope

Professor Imtiaz advised me to begin reading and prepare preliminary results or figures before scheduling a meeting. This package responds with a narrowed technical scope, a first experiment that can produce reviewable figures, and explicit decision gates before implementation expands.

Included in the proposed study	Excluded from the proposed study
One leader and one follower as the minimum validated system	Physical robot deployment during this course
Namespace-safe design that can launch additional followers	Global mapping, SLAM, GPS, or centralized fleet control
RGB detection, relative state estimation, EKF, PID, and recovery logic	LiDAR and stereo fusion from the earlier concept
Gazebo experiments, repeatable datasets, plots, and resource benchmarks	Claims of real-world safety or production certification

3 Research Questions and Hypotheses

Primary research question.

How accurately and reliably can a resource-constrained follower robot maintain a leader-relative formation using only monocular RGB detections and local odometry under changes in range, heading, velocity, and short visual occlusion?

Hypothesis	Testable claim	Evidence
H1 Relative perception	A calibrated monocular model will estimate leader range and bearing with bounded error over the declared operating envelope.	Range RMSE, bearing MAE, bias, and 95th-percentile error against Gazebo ground truth
H2 State estimation	An EKF will reduce tracking error and bridge short detection gaps more effectively than raw measurements.	Paired trials comparing raw, filtered, and prediction-only intervals
H3 Edge optimization	Reduced-precision inference will lower latency and memory use without materially degrading closed-loop tracking.	FP32, FP16, and INT8 detector and closed-loop benchmarks

4 Proposed Architecture

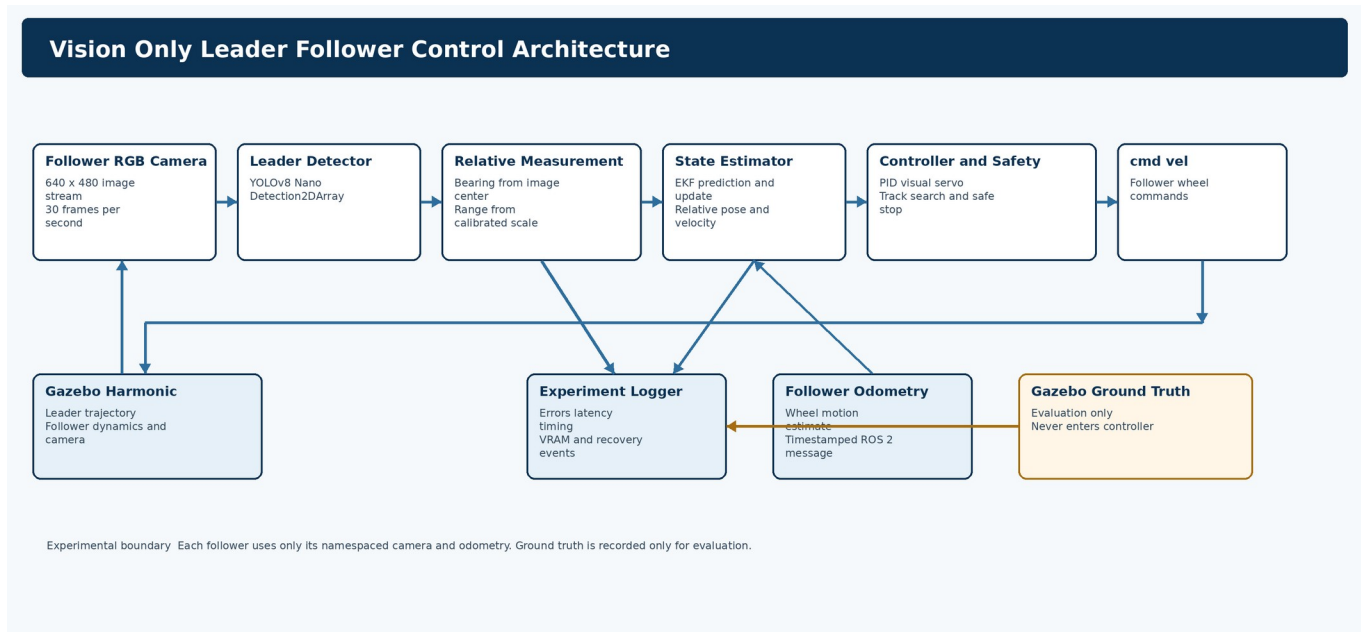


Figure 1 Proposed ROS 2 computational architecture and experimental boundary

COMPONENT RESPONSIBILITIES

Component	Input and responsibility	Output	Failure behavior
Camera bridge	Gazebo RGB frames with timestamps	sensor_msgs Image	Marks stale frames and exposes drop counts
Leader detector	Image stream; detect the leader rear profile	vision_msgs Detection2DArray	Publishes empty detection with timestamp
Relative measurement	Bounding box and camera calibration	Range, bearing, covariance	Rejects invalid geometry and low confidence
EKF estimator	Visual measurement and follower odometry	Relative pose, velocity, covariance	Prediction-only mode for bounded gaps
Supervisor	Detection health, covariance, and timeouts	TRACK, SEARCH, or SAFE STOP state	Commands safe stop when uncertainty exceeds limit
PID controller	Filtered range and bearing errors	geometry_msgs Twist	Velocity and acceleration limits remain active
Experiment logger	Ground truth and all pipeline telemetry	Versioned CSV and rosbag	Fails the trial if required evidence is missing

TIMING AND ISOLATION

Camera measurements are expected at up to 30 Hz, while the estimator and controller will run at a nominal 50 Hz using predictions between images. Separate callback groups and measured deadline statistics will isolate control work from variable detector latency. A `MultiThreadedExecutor` enables concurrency but does not by itself guarantee real-time scheduling. Each follower will consume only topics within its namespace; an automated graph audit will flag cross-namespace dependencies.

5 Preliminary Experiment for the First Faculty Meeting

The first experiment will validate the measurement model before the detector, EKF, and controller are integrated. A Gazebo leader model with known dimensions will be placed at controlled ranges and headings. The follower camera will record image geometry while Gazebo publishes ground truth to the evaluation logger only.

Element	Definition
Independent variables	Range 0.75 to 3.00 m; relative heading -30 to +30 degrees; three lighting levels; centered and off-axis placement
Measurements	Bounding-box center, width, height, area, timestamp, camera intrinsics, and Gazebo relative pose
Baselines	Analytic pinhole-inspired scale model and a fitted regression using held-out calibration poses
Controls	Fixed target model, camera resolution, focal parameters, random seed, and simulation step size
Repetitions	At least five seeded trials per condition after a short deterministic smoke-test grid
Outputs	Tidy CSV, configuration manifest, plots, summary statistics, and a reproducible launch command

PLANNED PRELIMINARY FIGURES

1. Estimated range versus Gazebo ground-truth range with an identity line and 95 percent confidence intervals.
2. Range residuals by distance and heading to reveal calibration bias and field-of-view effects.
3. End-to-end measurement latency and peak memory use for the baseline pipeline.
4. Example camera frames showing valid, marginal, and rejected bounding-box geometry.

DECISION RULE

The project proceeds to learned detection only if the calibrated geometry is identifiable and repeatable over the operating envelope. If range error grows beyond the proposed bound at oblique views, the scope will either narrow the allowable heading or replace box area with a target-specific apparent-width model. This decision will be documented before controller development.

6 Evaluation Plan and Proposed Targets

The values below are proposed engineering targets for discussion. They are not measured results and may be revised after the preliminary baseline.

Measure	Proposed target	Evaluation window
Relative range error	RMSE at or below 0.20 m	0.75 to 3.00 m operating range
Relative bearing error	Mean absolute error at or below 5 degrees	-30 to +30 degrees
Nominal formation error	Distance RMSE at or below 0.25 m and yaw MAE at or below 7 degrees	Straight and S-curve trajectories
Perception timing	At least 15 Hz and p95 latency at or below 67 ms	Gazebo and detector running concurrently
Control timing	50 Hz nominal with fewer than 1 percent missed periods	Each complete trajectory trial
Occlusion recovery	Median reacquisition at or below 3 s	0.5 to 1.5 s partial occlusions
Safety	Zero collisions in the declared test matrix	Minimum 30 seeded closed-loop trials
Resource use	No OOM; peak GPU memory at or below 3.6 GiB	15-minute integrated stress run

EXPERIMENTAL COMPARISONS

Factor	Levels
Estimator	Raw measurement; EKF; EKF with prediction during dropout
Inference precision	Reference FP32; TensorRT FP16; TensorRT INT8 if supported and accurate
Trajectory	Static; linear acceleration; sharp turn; S-curve; partial occlusion
Perception disturbance	Nominal; motion blur; reduced light; dropped frames

7 Evidence and Data Contracts

Every trial will be reproducible from a configuration file and seed. The evaluator will record synchronized perception, estimation, command, ground-truth, and resource data. Ground truth remains outside the controller so evaluation cannot leak privileged information into the autonomy stack.

Field group	Required contents	Purpose
Trial identity	run_id, git_commit, configuration_hash, seed, start_time	Reproduce and audit each result
Perception	frame_stamp, confidence, box center, width, height, inference_ms	Evaluate detection and timing
Estimation	range, bearing, velocities, covariance, estimator_mode	Compare estimates with ground truth
Control	state, distance_error, yaw_error, linear_cmd, angular_cmd	Explain closed-loop behavior
Ground truth	relative pose and relative velocity	Compute accuracy metrics only
Resources	CPU, RAM, GPU memory, dropped frames, missed periods	Verify edge constraints

8 Hardware and Implementation Constraints

Constraint	Observed baseline	Design response
Host	MSI Katana GF66 12UD; Intel Core i5-12450H; 16 GB RAM	Keep simulation-only scope and profile CPU callback timing
GPU	RTX 3050 Ti Laptop GPU with 4 GB VRAM	Small detector, reduced precision, bounded image size, headless benchmark mode
Storage	Approximately 294 GB free on NVMe	Retain selected rosbags; store derived CSVs and manifests in Git
Software	Ubuntu 24.04; partial ROS 2 Jazzy installation; CUDA 12.0	Create a pinned setup script and validate GPU access before ML integration
Current gap	ROS 2 CLI, Gazebo Harmonic, and inference runtimes are not yet available in the checked environment	Complete an environment smoke test as the first approved implementation milestone

The implementation will maintain a CPU-compatible evaluation path where practical. This prevents a GPU driver or TensorRT installation problem from blocking data analysis and lets the optimized backend be compared with a reference backend.

9 Milestones and Approval Gates

Each implementation milestone begins only after Neeraj approves its scope, acceptance criteria, tests, and documentation changes. Professor confirmation is required before the project moves from preliminary evidence to the full integrated research plan.

Dates	Milestone and deliverable	Exit evidence
Sep 14 to Sep 20	M0 Scope, architecture, environment, and reproducibility baseline	Professor package; clean ROS and Gazebo smoke test; system diagnostic report
Sep 21 to Oct 4	M1 Simulation world and calibrated relative-measurement experiment	Ground-truth comparison CSV and first four figures
Oct 5 to Oct 18	M2 Learned detector and EKF estimator	Held-out detector metrics; raw versus filtered ablation
Oct 19 to Oct 31	M3 Integrated controller, recovery states, and beta demonstration	Repeatable nominal and occlusion trials; October 31 integration review
Nov 1 to Nov 15	M4 FP16 and INT8 optimization and adversarial evaluation	Latency, accuracy, memory, and closed-loop comparison
Nov 16 to Nov 29	M5 Repeat experiments, thesis draft, and demonstration video	Frozen dataset; reproducible plots; complete draft and recorded demo
Nov 30 to Dec 4	M6 Final corrections and submission	Faculty-requested revisions incorporated; archived release

10 Risks and Mitigations

Risk	Why it matters	Mitigation and decision point
Scope ambiguity	The earlier LiDAR and stereo negotiation idea differs from the vision-only document.	Ask Professor Imtiaz to confirm the revised research question before full implementation.
The word swarm overstates validation	A single leader-follower pair does not demonstrate swarm behavior.	Use coordination in the title; make multi-follower support an extension after the pair is validated.
Monocular range bias	Bounding-box scale varies with heading, detection jitter, and partial visibility.	Calibrate first; quantify residuals; narrow the envelope or change the measurement model if needed.
Field-of-view loss	Sharp turns can break the feedback loop.	Use confidence and covariance gates, bounded prediction, SEARCH, and SAFE STOP states.
GPU memory exhaustion	Gazebo rendering and inference share 4 GB VRAM.	Start with FP16, record peak memory, use headless trials, then evaluate INT8.
Timing claims	A multi-threaded executor does not guarantee a strict 50 Hz loop.	Measure callback periods and missed deadlines; avoid unverified real-time claims.
Synthetic-only evidence	Simulator accuracy may not transfer to physical robots.	State the limitation; vary lighting, blur, occlusion, and model appearance; exclude deployment claims.
Schedule compression	Integration failures could consume the thesis-writing period.	Freeze the integrated beta on October 31 and protect November for experiments and writing.

11 Related Work and Technical Foundations

Source	Relevant contribution	Use in this project
Hutchinson, Hager, and Corke, 1996	Foundational visual-servo control framework	Frames image-derived error as feedback for robot motion
Vaitheeswaran et al., 2015	Camera-guided leader-follower ground vehicles	Provides a direct historical baseline for vision-only following
Bateux et al., 2017	Learning-based visual servoing with synthetic perturbations	Supports simulator-generated training and occlusion experiments
Santjoko et al., 2026	Leader-follower control with explicit field-of-view safety	Motivates treating visibility as a control and recovery constraint
ROS 2 and Gazebo documentation	Supported Jazzy and Harmonic integration with bridge interfaces	Defines the reproducible simulation and message boundary
NVIDIA TensorRT documentation	Explicit reduced-precision inference and accuracy tradeoffs	Defines the FP16 and INT8 benchmark methodology

12 Decisions Requested from Professor Imtiaz

1. Is the revised vision-only, simulation-only scope appropriate for the four-credit EE 616 project?
2. Should the required validation remain one leader and one follower, or should two followers be required to justify the term swarm?
3. Are the proposed preliminary figures sufficient for the first progress meeting?
4. Are there department-specific thesis, demonstration, or repository requirements that should be added now?

Recommended Decision and Immediate Next Step

Approve the revised scope for a preliminary baseline only. The immediate technical task is the calibrated monocular range-and-bearing experiment in Gazebo. Full detector, EKF, and controller implementation should follow after the first evidence is reviewed and the project scope is confirmed.

Appendix A Draft Email to Professor Imtiaz

Subject: EE 616 revised project scope and preliminary results plan

Hello Professor Imtiaz,

Following your recommendation, I have continued reading and refined the project into a simulation-focused study of vision-based leader-follower robot coordination on constrained edge hardware.

The revised scope uses ROS 2 Jazzy and Gazebo Harmonic. A follower robot will estimate the relative distance and heading of a leader using only an RGB camera and its own wheel odometry. An extended Kalman filter will smooth the relative-state estimate, and a visual-servo controller will maintain the desired formation and handle short visual occlusions. Gazebo ground truth will be used only to evaluate the estimates and controller.

This scope is narrower than my earlier LiDAR and stereo intersection-negotiation proposal. The narrower study gives me a clearer research question and makes it possible to produce controlled, repeatable results within the project period.

For the preliminary evidence, I plan to prepare:

1. A system architecture and ROS 2 computational-graph figure.
2. A comparison of monocular range and bearing estimates with Gazebo ground truth.
3. Initial perception latency and GPU memory measurements on my laptop.
4. A proposed matrix for tracking, sharp-turn, and occlusion-recovery experiments.

Could you please confirm whether this revised scope is appropriate for EE 616 before I proceed with the full implementation? I have attached the project update package for your review.

Regards,

Neeraj Kumar Kanchani

Appendix B References

- [1] Open Robotics. Gazebo Harmonic feature comparison and ROS 2 Jazzy integration. <https://gazebosim.org/docs/harmonic/comparison/>
- [2] Open Robotics. Using ROS 2 to interact with Gazebo. https://gazebosim.org/docs/harmonic/ros2_integration/
- [3] ROS 2 vision_msgs documentation for standardized detector messages. https://docs.ros.org/en/jazzy/p/vision_msgs/
- [4] NVIDIA. TensorRT quantization schemes. <https://docs.nvidia.com/deeplearning/tensorrt/latest/inference-library/quantized-types-schemes.html>
- [5] S. Hutchinson, G. D. Hager, and P. I. Corke. A Tutorial on Visual Servo Control. IEEE Transactions on Robotics and Automation, 12(5), 1996. <https://doi.org/10.1109/70.538972>
- [6] S. M. Vaitheeswaran, Bharath M. K., and Gokul M. Leader Follower Formation Control of Ground Vehicles Using Camshift Based Guidance, 2015. <https://arxiv.org/abs/1501.01364>
- [7] Q. Bateux, E. Marchand, J. Leitner, F. Chaumette, and P. Corke. Visual Servoing from Deep Neural Networks, 2017. <https://arxiv.org/abs/1705.08940>
- [8] I. R. Santjoko, R. R. Suganda, M. Pan, and B. Hu. Distributed 3D Leader-Follower Formation Control with Field-of-View Safety via Control Barrier Functions, 2026. <https://arxiv.org/abs/2605.17533>